

480ps 64-bit Race Logic Adder

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Abstract

In this paper, a high-speed 64-bit carry look-ahead adder is implemented by Race Logic for fast carry generation. G^1g/G^1k (Level 1 Group Generate/Kill) and G^2g/G^2k (Level 2 Group Generate/Kill) stages are designed by Race Logic. The adder consists of 4-stages, and clk-S63 delay is 480ps with 0.18 μ m CMOS technology.

Introduction

This paper presents a fast 64-bit carry look-ahead adder implemented by Race Logic for fast carry generation [1]. Race Logic can reduce the complexity of the circuit implementation of the given Boolean function, so that the delay of n-bit grouping of Generate/Kill is reduced to that of n-bit wired-OR gate which has simple structure and fast operation.

As the clock frequency of high performance micro-processors goes beyond 1GHz, requirements of high-speed adder became more stringent. The carry look-ahead structure for carry generation and carry select for sum generation is popularly adopted in a number of high-speed adders [2]. In the structures adopting carry look-ahead adder architecture, fast generation of Gg, Gp, and Gk is critical to reduce latency of the adder. Various circuit design styles have been presented for effective carry generation [2][3].

Overall Architecture

Figure 1 shows the overall architecture of 64-bit Race Logic adder. It consists of 4-stages: g/k generation, G^1g/G^1k generation, G^2g/G^2k generation, and SUM selection. The critical path of the adder is drawn by solid arrow. Until the G^2g/G^2k generates carries for MUX select, two 8-bit adders calculate SUMs for carry-0 and 1, respectively. The adders are carry look-ahead adders. In G^1g/G^1k stage, Group g/Group k for eight g/ks are generated. In G^2g/G^2k stage, the carries for 8-bit SUM carry-select, i.e. c7, c15, c23, c31, c39, c47, and c55 are generated simultaneously. The g/k generation circuit is designed by conventional dynamic logic. The MUX in the final stage is implemented by conventional pass gate logic. For the fast carry generation, Race Logic is used in G^1g/G^1k and G^2g/G^2k stages.

Fast Carry Generation Scheme with Race Logic

A. G^1g/G^1k generation circuit

In logic operation, Race Logic utilize timing difference between input variables instead of transistor logics [1]. Therefore, with Race Logic, the latency of Gg/Gk generation is not limited by transistor delay. This fact makes it possible to group 8-bit g/k for level-1 G^1g/G^1k generation, as shown

in equation 1, instead of generally adopted 4-bit grouping.

$$G^1g = g_7 + \overline{k_7}\{g_6 + \overline{k_6}\{g_5 + \overline{k_5}\{\dots\{g_1 + \overline{k_1}g_0\}\}\}\}\} \quad (\text{Eq. 1})$$

The figure 2 shows the circuit schematics of the G^1g/G^1k generation circuit designed by Race Logic. The circuit consists of three parts: clock distribution resistor array, input switches with two cascaded NMOS, and one pair of race lines (G-line and K-line). As shown in the figure 2, implemented G^1g/G^1k generation circuit is equal to 8-bit wired-OR gates.

When clk is low, the G-line and K-line are precharged high. The OUT and OUTb are also precharged high, so that outputs, G^1g and G^1k , are initially '0'. When the clk signal goes high, the input switches are sequentially triggered. If the data of a switch is '1', the switch will be turned on so that the race line to which the switch is attached will be discharged to '0'. On the other hand, if the data is '0', the switch does not make any effect on the race line. The Winner-Take-All-Circuit (WTAC) detects which race line falls down earlier. The OUT and OUTb are precharged low. If the G-line becomes '0' earlier than K-line, the OUT of the WTAC becomes '0', and the OUTb remains '1'. For example, in the case of $\{g_7, \dots, g_0\} = \{0, 0, 1, 0, 1, 0, 0, 1\}$ and $\{k_7, \dots, k_0\} = \{0, 0, 0, 1, 0, 0, 0, 0\}$, g_5 is triggered earlier than k_4 so that the G-line falls first, which makes OUT be '0' and G^1g be '1'. The timing diagram for this operation is illustrated in figure 3.

The circuit implementation of WTAC is shown in figure 4. When the clk is low, P_1 , P_2 , P_3 , and P_4 precharge G-line, OUT, K-line, and OUTb, respectively. N_G and N_K are initially turned on. If the clk goes high and the G-line falls earlier than K-line, OUT falls down consequently. As the OUT falls down, N_K is turned off and P_K is turned on, and OUTb is fixed to high state. Because the OUTb is connected to the gate of N_G again, N_G continues to connect G-line to OUT. By this feedback effect, the WTAC operates very fast, and can detect a very small timing gap between G-line and K-line signal. Furthermore, there is only one pass gate transistor between G-line and OUT. The simulation results of WTAC are shown in figure 5. The circuit simulation is performed by 0.18 μ m CMOS technology model parameter with the condition of 1.62Vdd and 85°C.

B. G^2g/G^2k generation circuit

In G^2g/G^2k stage, intermediate carries, i.e. c7, c15, c23, c31, c39, c47, and c55, are simultaneously generated for MUX select. Figure 6 shows the G^2g/G^2k generation block designed by Race Logic. Seven G/K-line pairs, WTACs, and input switches are used. N-switches are connected to the n-th G/K-line, and these switches have junction capacitance. Therefore, the capacitance of 7th G/K-line pair is the largest among G/K-lines, so that the generation of c_{55} is the critical

path in G^2g/G^2k stage.

Simulation Results

The simulation result for the worst input case are shown in figure 7. The delay times consumed in G^1g/G^1k and G^2g/G^2k stage are 185ps and 218ps, respectively. The layout of the adder is shown in figure 8. The implemented adder occupies $490\mu\text{m} \times 180\mu\text{m}$, and its transistor count is 4,307. As shown in the figure 8, the G^1g/G^1k and G^2g/G^2k generation blocks designed by Race Logics occupy very small area, only 20% of total area. The detailed delay time and area are summarized in table 1.

Conclusions

A fast 64-bit carry look-ahead adder is realized with Race Logic. The Race Logic reduced the complexity of conventional Gg/Gk generation circuit to that of simple wired-OR gate. G^2g/G^2k stage generates all carries for MUX select at the same time. The structure of the circuit is quite regular so that it occupies small area. The clk-SUM63 was 480ps using $0.18\mu\text{m}$ CMOS technology.

Reference

- [1] S. J. Lee and H. J. Yoo, "A Novel High Speed Low Power Logic Family : Race Logic," Proc. IEEE European Solid-State Circuit Conference, pp. 420-423, Sept. 2000.
- [2] J. Park, H. C. Ngo, J. A. Silberman, and Sang H. Dhong, "470ps 64bit Parallel Binary Adder," Proc. IEEE Symposium On VLSI Circuits, pp. 192-193, June, 2000.
- [3] K. Mashiko et al., "A 64-bit Carry Look Ahead Adder Using Pass Transistor BiCMOS Gates," IEEE Journal of Solid-State Circuits, vol. 31, no. 6, June, 1996.

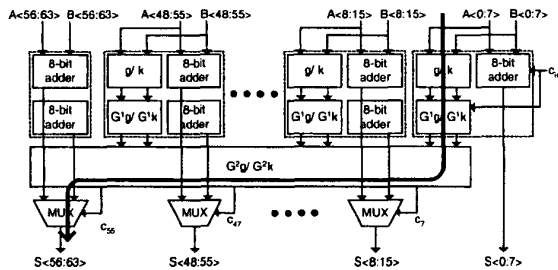


Fig. 1 Overall architecture

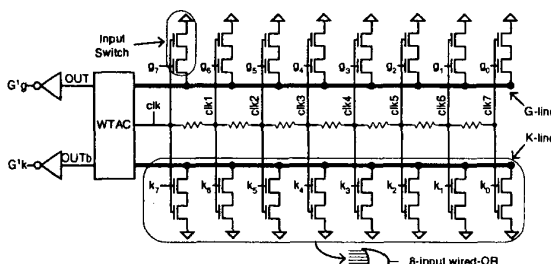


Fig. 2 G^1g/G^1k generation circuit implemented by Race Logic

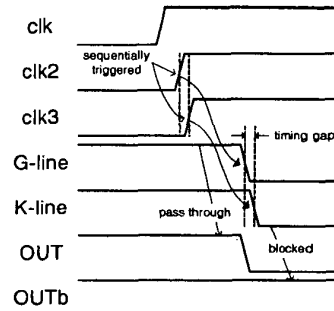


Fig. 3 Timing diagram of G^1g/G^1k generation circuit

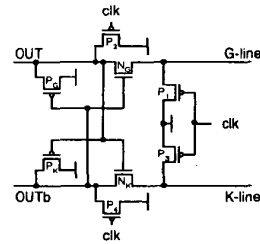


Fig. 4 Winner-Take-All-Circuit

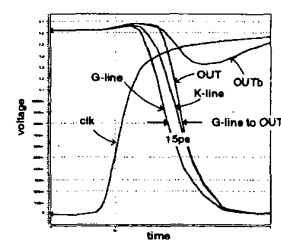


Fig. 5 Simulation result

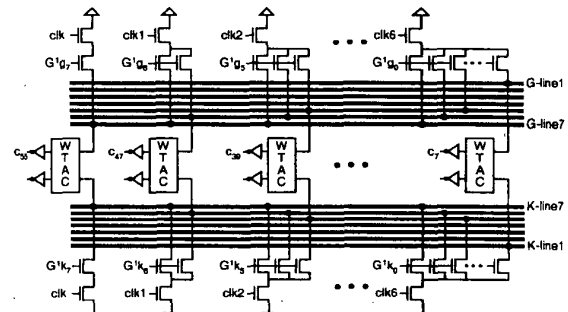


Fig. 6 G^2g/G^2k generation circuit using Race Logic

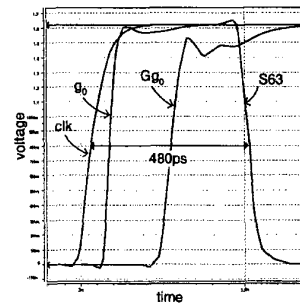


Fig. 7 Clk-SUM63 delay simulation result

Block	Delay (ps)	Area (%)
8-bit Adder	203	54%
g/k	57	20%
G^1g/G^1k	185	10%
G^2g/G^2k	218	10%
MUX	20	6%

Table 1 Summary of delay time and Area portion of the proposed adder

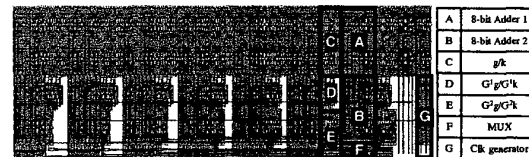


Fig. 8 Layout of the implemented adder